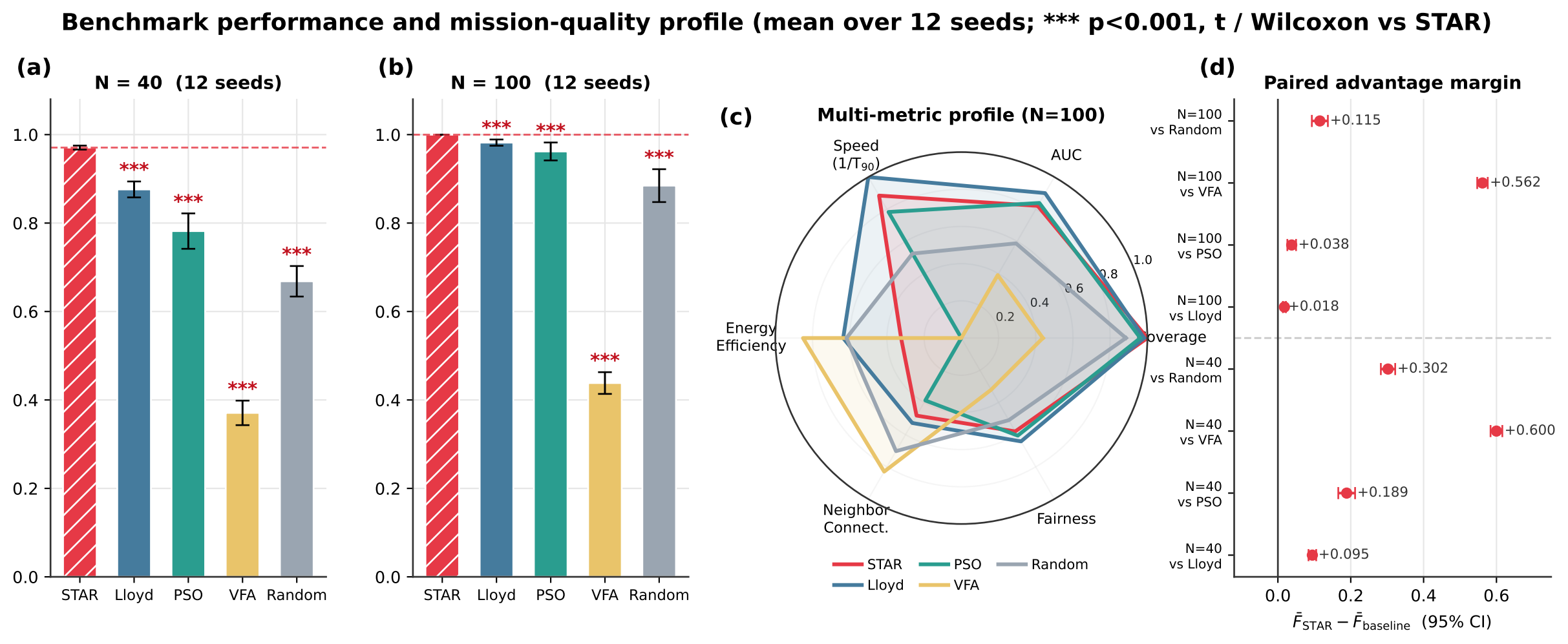
Benchmark performance and mission-quality profile (mean over 12 seeds; *** p<0.001, t / Wilcoxon vs STAR)
(a)
N = 40 (12 seeds)
(b)
N = 100 (12 seeds)
STAR
Lloyd
PSO
VFA
Random
***
(c)
Multi-metric profile (N=100)
Speed (1/T90)
AUC
Energy Efficiency
Neighbor Connect.
Fairness
Coverage
0.2
0.4
0.6
0.8
1.0
(d)
Paired advantage margin
N=100 vs Random
N=100 vs VFA
N=100 vs PSO
N=100 vs Lloyd
N=40 vs Random
N=40 vs VFA
N=40 vs PSO
N=40 vs Lloyd
+0.115
+0.562
+0.038
+0.018
+0.302
+0.600
+0.189
+0.095
$\bar{F}_{STAR} - \bar{F}_{baseline}$ (95% CI)